

VOLUME III - TECHNOLOGY / DOCUMENT 2 OF 4

Sensor Research

Positioning, Thermal, LiDAR, Visibility and Supporting Sensor Subsystems

Document ID	NMDC-LVH-TC-02
Volume	III - Technology
Programme	Safe & Continuous Haulage under Low-Visibility Monsoon Conditions
Site of Application	Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh
Classification	Internal Research / Solution Design Input
Revision	v1.0
Issue Date	20 September 2026

ABSTRACT

This document covers the sensing subsystems other than radar. The positioning stack receives the most detailed treatment, because precise positioning against a digital road model is the mechanism by which the architecture avoids needing to see the static world at all. The document then covers thermal imaging, LiDAR in its correct role as a mapping rather than a perception tool, visibility instrumentation, inertial and odometric sensing, and the sensor fusion framework that binds them. It concludes with the digital road model - treated here as a sensing asset, because in this architecture it substitutes for sensing.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. The Positioning Stack

1.1 Why positioning is the centre of the architecture

In a conventional perception architecture, positioning is a supporting function - useful for mapping and dispatch, secondary to the sensors that perceive the world. In the architecture proposed in Volume II Document 3, this is inverted.

If the vehicle's position is known to 0.1 m and the road is known to equivalent accuracy, then the road ahead can be drawn for the operator **with no forward sensing whatsoever**. The curve, the gradient, the berm, the junction, the tipping edge - all of it is in the map. Fog is irrelevant to a map.

This is why positioning receives more attention in this document than any other sensor: **it is the layer that converts an unsolvable perception problem into a solvable one**. Every metre of positional error is a metre of lane guidance uncertainty, and lane width margins on a Bailadila haul road are of the order of 1-2 m.

1.2 GNSS fundamentals and error sources

A GNSS receiver computes position by measuring pseudoranges to multiple satellites. Available constellations include GPS, GLONASS, Galileo, BeiDou and - of particular relevance in India - **NavIC/IRNSS**, with GAGAN providing satellite-based augmentation over the Indian region.

Error sources, with their behaviour under differential correction:

Error source	Magnitude (uncorrected)	Spatially correlated?	Removed by RTK?
Satellite clock	1-2 m	Yes	Yes
Satellite ephemeris	1-2 m	Yes	Yes
Ionospheric delay	2-10 m, variable	Yes over short baselines	Yes, over baselines under ~20 km
Tropospheric delay	0.2-1 m	Yes	Largely
Receiver noise	0.1-1 m	No	No, but small
Multipath	0.5 to several m	No	No - this is the dominant residual in a pit
Satellite geometry (DOP)	Multiplies all of the above	n/a	Geometry is what it is

Multipath is the critical entry. It is not spatially correlated between the rover and the base, so differential correction does not remove it. In a Bailadila pit, with high benched rock faces of highly conductive ore and large steel vehicle bodies in close proximity, multipath is the error source that determines achievable accuracy. This has direct design consequences discussed in Section 1.4.

1.3 Differential and RTK correction

Technique	Principle	Accuracy	Latency to fix	Infrastructure
SBAS (GAGAN)	Wide-area corrections broadcast from geostationary satellites	1-3 m	Immediate	None local
DGPS	Code-phase corrections from a local base	0.3-1 m	Seconds	Local base
RTK	Carrier-phase ambiguity resolution against a local base	0.01-0.05 m	Seconds to tens of seconds	Local base, short baseline
Network RTK	Corrections interpolated from a CORS network	0.02-0.05 m	Seconds	Network plus connectivity

Technique	Principle	Accuracy	Latency to fix	Infrastructure
PPP	Precise orbit and clock products, single receiver	0.05-0.1 m	Tens of minutes convergence	Correction service
PPP-RTK	PPP with regional atmospheric corrections	0.05 m	Faster than PPP	Regional network

Recommendation: RTK against local base stations. The baselines at Bailadila are short - the complexes are compact relative to RTK's effective range - which places the system in RTK's strongest regime. PPP is rejected because its convergence time is incompatible with intermittent operation: a truck emerging from a GNSS-shadowed section cannot wait tens of minutes for a solution.

Base station siting. Multiple base stations are likely required to cover both complexes with adequate geometry and to provide redundancy. Bases must be on stable, surveyed monuments outside the zone of mining-induced movement, with clear sky view, and with correction delivery over the site RF network. Base station integrity monitoring is essential - an undetected base station coordinate shift would introduce a systematic error into every vehicle simultaneously, which is a single point of failure requiring explicit mitigation.

1.4 The Bailadila GNSS environment

Bailadila presents a specifically difficult GNSS environment, and this should be assessed empirically in Phase 0 rather than assumed.

- ? **Sky obstruction.** Working in a pit or against a highwall removes a substantial fraction of the sky. Satellite count falls, DOP rises, and the solution degrades or is lost.
- ? **Switchbacks against rock faces.** A truck on a hairpin has a rock face on one side and open space on the other, giving asymmetric obstruction that changes rapidly through the turn.
- ? **Multipath from ore faces.** High-grade haematite is highly conductive. Reflections from benched ore faces at close range produce multipath that differential correction cannot remove.
- ? **Equatorial ionosphere.** India lies in the equatorial anomaly region, where ionospheric activity and scintillation are more pronounced than at mid-latitudes, occasionally degrading carrier-phase tracking. Short RTK baselines largely mitigate the delay component; scintillation-induced loss of lock is harder to mitigate and argues for robust reacquisition and strong inertial coasting.
- ? **Monsoon cloud.** Cloud does not affect GNSS signals; heavy rain has negligible effect at L-band. This is worth stating because it is occasionally assumed otherwise - **the same weather that destroys optical sensing leaves GNSS untouched**, which is precisely why this architecture works.

Phase 0 action. A GNSS availability and accuracy survey along the full haul road network, in both complexes, logging satellite visibility, DOP, fix type and position scatter against a surveyed reference. This produces a map of where RTK is available, where it degrades, and where inertial coasting must carry the solution - which in turn sizes the IMU requirement and identifies any sections needing supplementary positioning.

1.5 Inertial sensing and tight coupling

An inertial measurement unit provides angular rate and specific force at high rate, continuously and independently of any external signal. Fused with GNSS it delivers what neither provides alone: GNSS bounds the drift, and inertial provides continuity and high-rate attitude.

IMU grade	Gyro bias stability	Typical horizontal drift over 60 s outage	Relative cost	Assessment
Consumer MEMS	10-100 deg/hr	Tens of metres	Very low	Insufficient
Industrial MEMS	1-10 deg/hr	Several metres	Low	Marginal; adequate only for very short outages

IMU grade	Gyro bias stability	Typical horizontal drift over 60 s outage	Relative cost	Assessment
Tactical MEMS / FOG	0.1-1 deg/hr	Sub-metre to ~2 m	Moderate	Recommended
Navigation grade	< 0.01 deg/hr	Centimetres	Very high	Unnecessary for this application

Coupling architecture. Three options exist:

- ? **Loose coupling** fuses the GNSS position solution with the inertial solution. Simple, but requires GNSS to produce a full position fix - so it provides no benefit when fewer than four satellites are visible.
- ? **Tight coupling** fuses raw pseudorange and carrier-phase observations with inertial data. **This is the recommended approach**, because it extracts value from partial satellite visibility - two or three satellites still constrain the solution, which is exactly the situation on a switchback against a rock face.
- ? **Deep coupling** feeds the inertial solution back into the receiver tracking loops, improving signal tracking under dynamics and scintillation. Higher complexity; worth consideration given equatorial scintillation risk but not essential.

Recommendation. Tight coupling with a tactical-grade MEMS IMU, with the outage duration requirement (PR-09, no gap exceeding 2 s) sized against the GNSS availability survey of Section 1.4. Where the survey reveals outages materially longer than the IMU can bridge, radar odometry aiding (Volume III Document 1, Section 8) and berm tracking supply the additional constraint.

1.6 Dual-antenna heading

Heading accuracy of better than 1 degree is required by PR-10. Three approaches:

Method	Accuracy	Viability at Bailadila
Magnetometer	1-3 deg nominal	Rejected. High-grade haematite ore body plus massive steel vehicle bodies make magnetic heading unreliable. Constraint C-06.
Course over ground from GNSS	Good when moving, undefined when stationary or reversing	Insufficient. A truck spotting at a face or reversing at a tip has no valid course-over-ground heading, and those are exactly the high-risk manoeuvres.
Dual-antenna GNSS (moving baseline)	0.1-0.3 deg with a 2 m baseline	Recommended. Valid when stationary, accurate, and immune to the ferrous environment.

The cost of the second antenna and receiver channel is modest and the capability is decisive: heading error translates directly into lateral error at range. At 80 m, a 1 degree heading error displaces a projected path by 1.4 m - more than the lane margin. Dual-antenna heading should be regarded as mandatory, not optional.

1.7 Odometry and non-holonomic constraints

Wheel speed and steering angle, available from the machine's own CAN bus on most modern HEMM, contribute two valuable constraints to the filter:

- ? **Odometric velocity**, which bounds inertial velocity drift during GNSS outage.
- ? **Non-holonomic constraints** - a wheeled vehicle does not move sideways or vertically relative to its own body frame - which suppress two of the three velocity error components almost entirely, substantially extending useful coasting duration.

Wheel slip on wet ore fines degrades odometry precisely during monsoon conditions, so odometry must be weighted adaptively, with slip detected by comparing odometric and inertial accelerations.

1.8 Integrity monitoring

Requirement PR-11 demands a real-time confidence figure, and requirement HR-11 demands that the interface degrade visibly. This requires integrity monitoring in the aviation sense: not merely an accuracy estimate, but a bounded confidence that the error does not exceed a stated limit.

- ? **RAIM / ARAIM.** Receiver autonomous integrity monitoring detects inconsistent satellite observations and, with sufficient redundancy, excludes the faulty one.
- ? **Protection levels.** Compute a horizontal protection level bounding the position error to a specified probability, and compare it against an alert limit derived from lane geometry. **When the protection level exceeds the lane margin, lane guidance must be withdrawn** - visibly, per HR-11.
- ? **Innovation monitoring** within the fusion filter, detecting inconsistency between GNSS, inertial, odometric and radar-odometric inputs.
- ? **Cross-validation against the road model.** If the position solution places the vehicle outside any known road, either the position or the map is wrong. Both cases warrant an alert.

The protection-level concept is the single most important idea transferred from aviation positioning practice. It replaces "the system thinks it is accurate" with "the system guarantees, to a stated probability, that the error is below this bound." Only the latter is a safe basis for guiding a loaded truck between berms in zero visibility.

1.9 Draft positioning specification

Parameter	Requirement	Traces to
Constellations	GPS, GLONASS, Galileo, BeiDou, NavIC; multi-frequency	Availability in a constrained sky
Correction	RTK from local base network; SBAS/GAGAN fallback	Section 1.3
Horizontal accuracy, open sky	≤ 0.10 m (95%)	PR-08 with margin
Horizontal accuracy, constrained sky	≤ 1.0 m (95%)	PR-08
Heading accuracy	≤ 0.3 deg, dual-antenna, valid at zero speed	PR-10
Attitude (pitch, roll)	≤ 0.1 deg	Radar ground plane estimation
Output rate	≥ 100 Hz fused solution	PR-12, PR-14
Outage bridging	≤ 0.5 m drift over 30 s; no solution gap exceeding 2 s	PR-09
IMU grade	Tactical MEMS, bias stability ≤ 1 deg/hr	Section 1.5
Coupling	Tight GNSS/INS with odometric and non-holonomic aiding	Section 1.5, 1.7
Integrity	Horizontal protection level output at ≥ 10 Hz; RAIM/ARAIM	PR-11
Aiding inputs	Wheel odometry, steering angle, radar odometry	Section 1.7, 1.8
Environmental	Per ER-01 to ER-05	ER series

2. The Digital Road Model

The road model is treated in this document because, in this architecture, **it performs the function of a sensor**: it supplies the static world that would otherwise have to be perceived.

Contents. Road centreline and edges with elevation; lane definition and direction of travel; gradient and curvature; design and zone speed limits; junctions, passing bays and single-lane sections; tipping edges and dump crests; loading areas; exclusion and geofenced zones; berm positions; fixed infrastructure; and for each element a validity timestamp and provenance.

Creation. Survey-grade LiDAR in clear conditions - mobile mapping from a survey vehicle or UAV-borne scanning - registered to the mine survey datum, is the appropriate tool. **This is LiDAR's correct role in this architecture: build the map in good weather, so that in bad weather no LiDAR is needed at all.**

Maintenance is the hard problem. Haul roads change continuously as ramps extend and benches advance (constraint C-13). Requirement FR-07 demands field update within one working shift without vendor involvement. Practical mechanisms:

- ? **Survey-integrated update.** Mine survey already records road development; the workflow should push these updates into the road model as a routine output rather than a separate task.
- ? **Fleet-sourced update.** Every equipped vehicle logs its precise trajectory. Systematic divergence of driven paths from the modelled road indicates a change. This is automatic, continuous, free, and should trigger a survey verification rather than an automatic model edit - the model is safety-critical and should not self-modify.
- ? **Field marking.** A supervisor with a handheld device marks a new section, a closure or a hazard, which propagates as a provisional model element flagged as unverified.
- ? **Staleness detection and display.** Any section whose validity timestamp exceeds tolerance, or against which driven-path divergence is detected, is rendered in a distinct degraded style with explicit warning (risk R4).

The road model is the architecture's principal new safety-critical asset and its principal new failure mode. A stale model in dense fog will confidently draw a road that is not there. Version control, validity timestamps, divergence monitoring and honest degraded rendering are not refinements - they are the controls that make the architecture safe.

3. Thermal Imaging

Technology. Uncooled LWIR microbolometer arrays, 8-14 μm , typically 640x480 or higher, 30-60 Hz, requiring no illumination and no cooling.

Genuine strengths for this application. Independent of illumination, so equally effective at night and unaffected by headlight backscatter - a meaningful advantage over visible cameras in fog, where backscatter is the dominant failure. Excellent contrast on thermally distinct targets: people, running engines, hot brakes and tyres, recently operated machinery. These are among the highest-consequence targets, and a person is one of the best thermal targets available. Penetrates dust and smoke substantially better than visible.

Honest limitations. The claim that thermal "sees through fog" requires qualification, and the qualification matters:

- ? At 10 μm wavelength against 10 μm droplets, the size parameter is approximately 3 - still within the Mie regime. Attenuation in dense fog is significant, not negligible.
- ? **High humidity degrades LWIR independently of droplets**, because water vapour absorbs within parts of the 8-14 μm band. Bailadila's monsoon combines the two.
- ? Thermally neutral targets are near-invisible: a cold rock on the road, a parked trailer that has stood for hours, a wet berm after rain, standing water. These are the monsoon hazards.

- ? The germanium window is expensive and **fouls with ore mud exactly like any other optical aperture** (constraint C-07).
- ? No direct range measurement. Range must be inferred from apparent size and known geometry, or supplied by fusion with radar.

Recommended role. A **secondary classification layer**, forward-facing, active in classes V0-V3, contributing:

- ? Pedestrian detection and confirmation, fused with radar micro-Doppler classification.
- ? Confirmation of radar tracks, improving classification confidence and thereby reducing false alerts (HR-06).
- ? Detection of hot spots indicating machinery, including a stalled machine with a running engine.
- ? Night operation in clear and light-haze conditions, where it materially outperforms visible cameras.

It should **not** be presented to the operator as a fog-penetrating view, because at V4 it is not one, and presenting it as such directly creates the over-trust failure mode F5.

4. LiDAR - The Mapping Role

LiDAR is excluded as a real-time perception layer on physics grounds: published fusion studies report LiDAR alone failing beyond approximately 70 m in dense fog, with useful range collapsing far further at MOR 3-5 m, and with near-field droplet backscatter generating a dense false return cloud.

Its correct role is mapping. Survey-grade LiDAR operated in clear conditions is the best available tool for generating the digital road model - road geometry, berm profiles, bench edges, gradient and curvature - at the accuracy the architecture requires. Periodic re-scanning maintains model currency.

This inversion is worth stating explicitly because it is counter-intuitive: **the sensor that fails most completely in fog is the one that makes fog-independent guidance possible**, because it builds the map that replaces perception. LiDAR is used once, in good weather, and its product is used in all weather.

Secondary real-time role. A vehicle-mounted LiDAR would be valuable in classes V0-V2 for precise road-edge detection and obstacle geometry, and for continuous map verification. Whether this justifies the cost is a Phase 3 question. It should never be relied upon at V3-V4, and the fusion layer must down-weight it automatically as measured visibility falls - the adaptive weighting pattern demonstrated in the research literature, where fusion shifts weight toward radar as visibility decreases.

5. Visibility Instrumentation

Visibility measurement underpins requirements FR-01 to FR-04 and the acceptance regime AR-02.

Instrument	Principle	Accuracy	Cost	Role
Forward-scatter visibility meter	Measures scattered light at a fixed angle over a short baseline; infers extinction coefficient	Good across a wide range	Moderate	Primary network instrument. Compact, low power, solar-compatible, suitable for distributed roadside nodes
Transmissometer	Measures direct transmission over a fixed baseline	Highest accuracy, especially at low MOR	Higher; needs fixed baseline and alignment	Reference instrument. One or two installations for acceptance testing and network calibration
Camera-based estimation	Infers visibility from image contrast degradation	Moderate; needs calibration	Very low	Network densification using existing CCTV

Instrument	Principle	Accuracy	Cost	Role
Vehicle-derived estimation	Infers visibility from onboard camera or from radar-versus-camera detection disparity	Moderate	Zero marginal hardware	Phase 3. Every truck becomes a mobile sensor, giving dense coverage of a patchy phenomenon

Network design. Bailadila fog is stratified by elevation and patchy in the advection regime (Volume I Document 1, Section 3.2), so a single instrument is inadequate and would produce exactly the mine-wide-declaration problem the architecture seeks to remove. The network should include:

- ? Instruments at multiple elevation bands from pit bottom to highest working bench, in both complexes.
- ? Instruments at the specific locations where operating decisions are made: ramp heads, major junctions, crusher tips, plant approaches.
- ? Co-located meteorological sensing - temperature, dew point, relative humidity, wind speed and direction, rainfall - because these are the predictors from which the nowcasting model of FR-04 learns.
- ? A reference transmissometer against which the forward-scatter network is periodically verified.

Power. Solar-plus-battery with not less than 7 days autonomy at monsoon insolation (ER-08). This is a demanding requirement precisely because the system is most needed when solar yield is lowest - persistent monsoon cloud is the condition being measured. Battery sizing must be based on measured local insolation during monsoon, not on annual averages.

Deploy this layer first. It is the cheapest element of the architecture, has no dependencies on any other subsystem, immediately converts the operating-mode decision from subjective to instrumented, provides the independent measurement required for acceptance testing (AR-02), and begins accumulating the training data on which nowcasting skill depends. Every month of delay in deploying it is a month of lost training data.

6. Ultrasonic and Magnetic Short-Range Sensing

Ultrasonic. Acoustic, therefore unaffected by fog. Range limited to a few metres and degraded by wind and heavy rain. Appropriate only for very-short-range spotting and manoeuvring assistance. Not a haul road technology; included for completeness.

Magnetic field proximity detection. Generates a shaped low-frequency magnetic field around the machine and detects personal devices entering it. Properties valuable for the pedestrian layer: does not require line of sight, penetrates machine structure and intervening bodies, gives a stable and well-defined zone geometry, and is entirely unaffected by fog, dust or smoke.

Recommended role. The **pedestrian blind-zone layer**, complementing radar, where its non-line-of-sight property covers the case of a person directly behind or beneath a structure that radar cannot illuminate.

Bailadila-specific caution. The technology depends on predictable magnetic field geometry. A high-grade haematite ore body and massive steel machine bodies may distort that geometry in ways that are not present in the coal and hard-rock environments where the technology is most established. **Requirement ER-10 applies with particular force here**, and on-site validation should precede any commitment to this layer.

7. Supporting Vehicle Sensing

Sensor	Source	Use
Wheel speed	Machine CAN	Odometry; ego-velocity for radar clutter rejection

Sensor	Source	Use
Steering angle	Machine CAN	Path prediction; odometric constraint
Brake and retarder state	Machine CAN	Intent broadcast in V2V message; alert suppression when already braking
Transmission state and direction	Machine CAN	Reversing state broadcast; rear alerting priority
Payload / suspension pressure	Machine system	Load state affects braking performance and therefore alert thresholds (FR-24)
Dump body position	Machine system	Tipping state broadcast; already a DGMS-required indication
Engine and system health	Machine CAN	Breakdown state broadcast, enabling automatic hazard marking of a stalled machine
Operator alertness	Dedicated camera or steering-behaviour analysis	Complementary to this programme; noted as an existing capability in mining CAS suites

CAN integration note. Requirement ER-05 mandates galvanic isolation from machine networks, and any connection must be **read-only unless an intervention function is approved** (requirement FR-26 and GR-03). Reading machine state is low-risk and high-value; writing to machine networks is a different regulatory category entirely and must not be introduced incidentally.

Fleet heterogeneity (constraint C-20) means CAN availability and signal definitions vary by machine make, model and age. The design must degrade gracefully where CAN data is unavailable, deriving what it can from GNSS/INS - and this fallback path should be tested, not assumed.

8. Sensor Fusion Framework

Architecture. A two-level scheme is recommended.

- ? **Level 1 - Positioning fusion.** A tightly coupled GNSS/INS/odometry/radar-odometry filter producing position, velocity, attitude and an integrity bound at 100 Hz. Error-state Kalman filtering is the established approach.
- ? **Level 2 - Object fusion.** A multi-target tracking framework fusing radar detections, cooperative V2V reports, thermal and camera detections, and short-range sensor inputs into a single track list with identity, kinematics, class and confidence, at 10-20 Hz, satisfying FR-15 and FR-16.

Fusion strategy. Track-level fusion is recommended over raw-data fusion: each sensor produces detections or tracks in its own frame, which are then associated and fused. This is more robust to individual sensor failure, easier to validate, and allows sensors to be added or removed without re-architecting - important for a phased rollout across a heterogeneous fleet.

Adaptive weighting. The fusion layer must weight sensor contributions by their **currently valid** reliability, not by a fixed configuration. The research literature demonstrates exactly this pattern: an integrated architecture with a performance assessment layer that monitors LiDAR degradation and shifts reliance to complementary sensors as performance falls below a critical level, and an adaptive fusion kernel balancing LiDAR spatial precision against radar penetration as visibility decreases.

Applied here:

Condition	Radar	Cooperative RF	Thermal	Camera	Map + position
V0-V1 clear	High	High	Moderate	High	High
V2 moderate fog	High	High	Moderate	Low	High

Condition	Radar	Cooperative RF	Thermal	Camera	Map + position
V3 dense fog	High	High	Low	Zero	High
V4 severe fog	High	High	Very low	Zero	High
Radar blocked	Zero	High	As above	As above	High
GNSS degraded	High	High	As above	As above	Reduced; guidance withdrawn if protection level exceeds lane margin

Association against the map. A distinguishing feature of this architecture is that fusion occurs **against the road model**, not in free space. A track is associated with a road segment and a lane; its predicted trajectory follows the road rather than a straight line. This substantially improves prediction quality on a curved haul road and is central to satisfying requirement FR-19 - and therefore to controlling the nuisance alert rate that risk R3 identifies as the highest-likelihood failure.

9. Calibration and Maintenance

Item	Requirement	Frequency
Radar extrinsic alignment	Documented alignment at installation; online drift monitoring per RR-06	Install; continuous monitoring
Camera and thermal intrinsics	Factory; verified at installation	Install; on replacement
GNSS antenna lever arms	Surveyed at installation to centimetre accuracy	Install; after any structural repair
IMU-to-body alignment	Estimated online during motion	Continuous
Visibility meter calibration	Against reference transmissometer	Per manufacturer, minimum annually
RTK base coordinates	Surveyed; monitored for movement	Install; periodic verification
Road model currency	Validity timestamp; divergence monitoring	Continuous; survey update as roads change
Sensor aperture cleanliness	Pre-shift check integrated with DGMS checklist per RR-03	Every shift
Blockage self-test	Continuous automatic	Continuous

Calibration drift is failure mode F10 and it is insidious: the system continues to produce plausible output that is systematically wrong. A radar boresight drift of one degree misplaces an 80 m target by 1.4 m - enough to assign it to the wrong lane, which either raises a false alert or, worse, suppresses a true one. Online drift monitoring is not optional.

10. Consolidated Sensor Specification

Subsystem	Specification summary	Priority
GNSS/INS	Multi-constellation multi-frequency RTK, dual-antenna heading ≤ 0.3 deg, tactical MEMS IMU, tight coupling, odometric and radar-odometry aiding, 100 Hz output, protection level at ≥ 10 Hz	M
RTK infrastructure	Local base network covering both complexes, surveyed monuments, integrity monitored, corrections over site RF	M

Subsystem	Specification summary	Priority
Digital road model	Survey-grade LiDAR derived, registered to mine datum, field-updatable within a shift, validity timestamps, divergence monitoring, staleness rendering	M
Radar	Per Volume III Document 1, Section 11	M
Thermal	LWIR 8-14 um, >= 640x480, >= 30 Hz, IP67, active in V0-V3, secondary classification only	S
Camera	Forward and rear, IP69K, heated and wiped or air-purged, active V0-V2, rear-view per GR-02	M (rear), S (forward)
Visibility network	Forward-scatter meters at multiple elevation bands both complexes; reference transmissometer; co-located met sensing; 7-day solar autonomy at monsoon insolation	M
Magnetic PDS	Personnel blind-zone layer, subject to on-site ferrous validation per ER-10	S
Ultrasonic	Very-short-range manoeuvring assistance	C
Machine CAN interface	Read-only, galvanically isolated, graceful degradation where unavailable	S
Fusion	Two-level; track-level object fusion; visibility-adaptive weighting; association against the road model	M